

RGB three-color lights

1. Learning target

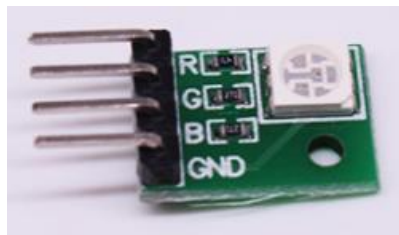
- 1.1 In this course, we will learn how to use pins of the Raspberry Pi Pico 2/Pico board.
- 1.2 How to control the RGB light module.

2. Preparation

Raspberry Pi Pico 2/Pico board *1
 Pico sensor expansion board *1
 PC *1
 USB data cable *1
 RGB light module *1
 male-to-male DuPont line *4

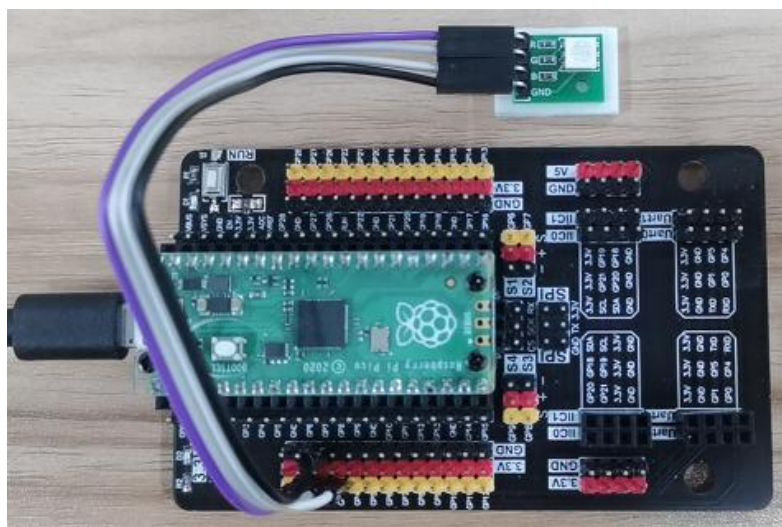
Note:

The RGB three-color light module is composed of three LEDs: red, green, and blue. When we provide high-level voltage to the R, G, and B pins, it will light up the corresponding color, and it is low-level voltage, RGB light will turn off.



3. About wiring

RGB light module	Pico sensor expansion board
R	GP1
G	GP2
B	GP3
GND	GND



4. About code

Thonny programming

About how to using ThonnyIDE, please check the tutorials in 【2.Development environment】

```
from machine import Pin
import utime

# Initialize the LED lights
red = Pin(1, Pin.OUT)
green = Pin(2, Pin.OUT)
blue = Pin(3, Pin.OUT)

# Control the red light, state=0 light is off, state= other values light is on
def rgb_red(state):
    if state == 0:
        red.value(0)
    else:
        red.value(1)

# Control the green light, state=0 light is off, state= other values light is on
def rgb_green(state):
    if state == 0:
        green.value(0)
    else:
        green.value(1)

# Control the blue light, state=0 light is off, state= other values light is on
def rgb_blue(state):
    if state == 0:
        blue.value(0)
    else:
        blue.value(1)

# Close RGB light
def rgb_off():
    red.value(0)
    green.value(0)
    blue.value(0)

# RGB light become white
def rgb_on():
    red.value(1)
    green.value(1)
    blue.value(1)
```

```
# Main loop, switch a color every 0.5s
```

```
while True:
```

```
    rgb_off()
```

```
    rgb_red(1)
```

```
    utime.sleep(.5)
```

```
    rgb_off()
```

```
    rgb_green(1)
```

```
    utime.sleep(.5)
```

```
    rgb_off()
```

```
    rgb_blue(1)
```

```
    utime.sleep(.5)
```

```
    rgb_on()
```

```
    utime.sleep(.5)
```

5. Phenomenon

Click the green run button  of Thonny IDE to start running the program. Click the red stop

button  to stop the program.

When the program is running, RGB three-color light will cyclically light up red, green, blue, and white, switching a color every 0.5 seconds.